

Article

Preparation of a Magnetically Recoverable Reed-Fiber Biosorbent for Cationic Dye Capture in Water: Equilibrium, Kinetics, and Reuse Behavior

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Abstract: A magnetically recoverable biosorbent was prepared from reed-fiber flour (RF) by depositing iron oxide domains through a mild aqueous co-precipitation route. The resulting RF/Fe₃O₄ material was designed for removal of a cationic blue dye from water while allowing fast recovery with an external magnet. Structural, magnetic, and adsorption tests indicated that the iron oxide phase was dispersed along the plant-fiber surface without destroying the fibrous backbone. Under neutral starting conditions, the material approached equilibrium within 45 min for a 30 mg L⁻¹ dye solution, and the uptake increased when the liquid phase became weakly alkaline. Equilibrium data were consistent with a Langmuir-type description, whereas the rate profile was best represented by a pseudo-second-order model. After six adsorption-regeneration cycles, the material retained most of its initial dye capture performance. The study outlines a low-energy pathway for producing a separable plant-fiber sorbent and provides a practical data set for designing small wastewater-polishing units.

Keywords: reed fiber; magnetic biosorbent; cationic dye; adsorption kinetics; regeneration; water polishing

1 Introduction

Synthetic dyes remain widely used in textile finishing, package printing, paper coating, plastics coloring, and several laboratory procedures. Even when the total mass released from a facility is modest, highly colored effluents can reduce light penetration, interfere with aquatic photosynthesis, and introduce persistent aromatic compounds into receiving waters [1]. Many cationic dyes also form transformation products that are more mobile than the parent colorant, which has encouraged the development of simple polishing methods that can be operated near the discharge point [2, 3].

Adsorption is attractive for small and medium effluent streams because the equipment is compact, start-up is rapid, and chemical dosing can be kept low [4]. Activated carbon remains a benchmark sorbent, but its preparation often requires high-temperature activation, extensive washing, or expensive feedstock selection [5]. Research has therefore shifted toward waste-derived biosorbents prepared from agricultural residues, aquatic plants, fruit peel, hulls, stems, and forestry by-products [6, 7]. These materials offer hydroxyl, carboxyl, phenolic, and ether groups that can interact with cationic dye molecules through electrostatic attraction, hydrogen bonding, and surface partitioning [8].

A common obstacle for fine biomass sorbents is separation after treatment. Filtration can clog, centrifugation increases energy use, and incomplete recovery can release solids into the treated water [9]. Magnetic modification is a practical strategy for solving this issue. Magnetite-bearing sorbents can be gathered quickly with a low-cost magnet and then transferred into a regeneration bath without complex solid-liquid equipment [10]. When the magnetic component is immobilized on a low-cost biomass carrier, the hybrid material combines accessible surface chemistry with easy handling [11].

Reed stems were selected in this work as a representative plant-fiber residue because they are abundant in wetland maintenance operations and possess a naturally channeled texture. The present article describes the preparation of RF/Fe₃O₄, its characterization, and its behavior during capture of a model cationic blue dye. The material is not intended as a finished commercial product; rather, it provides a format for evaluating how plant-fiber morphology, magnetic recovery, and adsorption performance can be combined in a single low-complexity system.

2 Results and Discussion

2.1 Material characterization

2.1.1 Phase features

Figure 1 presents the diffraction patterns of RF before and after alkaline washing, followed by the stacked patterns of RF/Fe₃O₄ prepared with increasing salt charge. The washed fiber retained the broad cellulose-related reflections typical of lignocellulosic feedstocks, while the relative intensity of the main cellulose reflection increased after soluble extractives were removed. For the magnetic materials, weak peaks associated with spinel iron oxide appeared progressively as the salt charge increased. The coexistence of the plant-fiber envelope and magnetite-related reflections indicates that the inorganic phase formed on the organic support rather than replacing it.

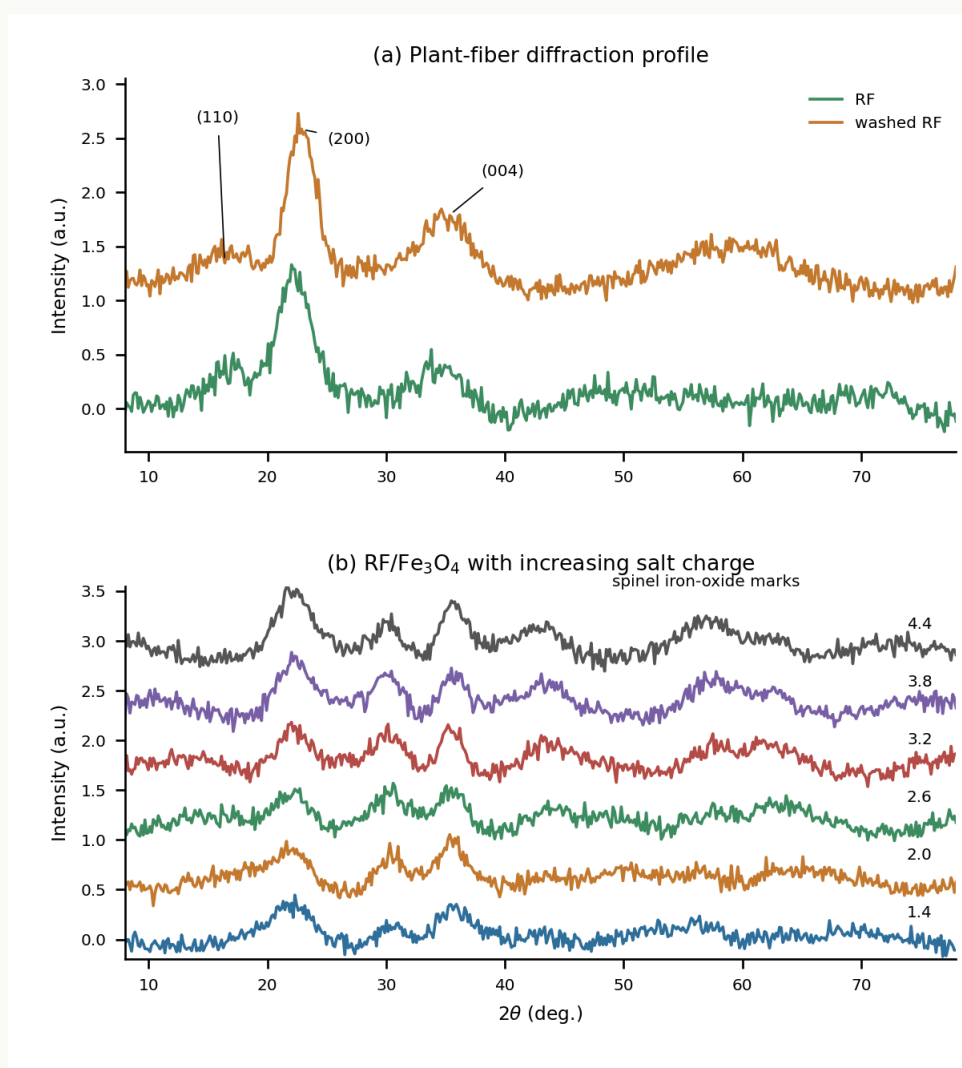


Figure 1: Diffractive signatures of the plant-fiber substrate and the magnetic composite. (a) RF before and after alkaline washing. (b) RF/Fe₃O₄ prepared with increasing iron-salt charge. Curves are shifted vertically for readability.

The increase in peak definition was moderate, which is consistent with nanoscale iron oxide domains dispersed on a rough support. A very sharp magnetite pattern would imply large crystallites and a lower active surface area. In contrast, the broad reflections observed here suggest that the inorganic phase remained finely divided, a condition favorable for both magnetic response and accessible surface interactions [12].

2.1.2 Morphology and elemental distribution

The replacement micrographs in Figure 2 show the visual progression from raw reed flour to alkali-washed fiber and then to the magnetic composite. The raw material appeared straw-colored with uneven surface debris. After washing, the surface became cleaner, and the fibrous grooves became easier to distinguish. The magnetic material showed darker surface patches attributed to deposited iron oxide. At higher magnification, the RF/Fe₃O₄ texture was more granular than the washed fiber, which agrees with a surface deposition mechanism. Elemental mapping also showed iron- and oxygen-rich regions distributed along the fiber rather than concentrated in one isolated cluster.

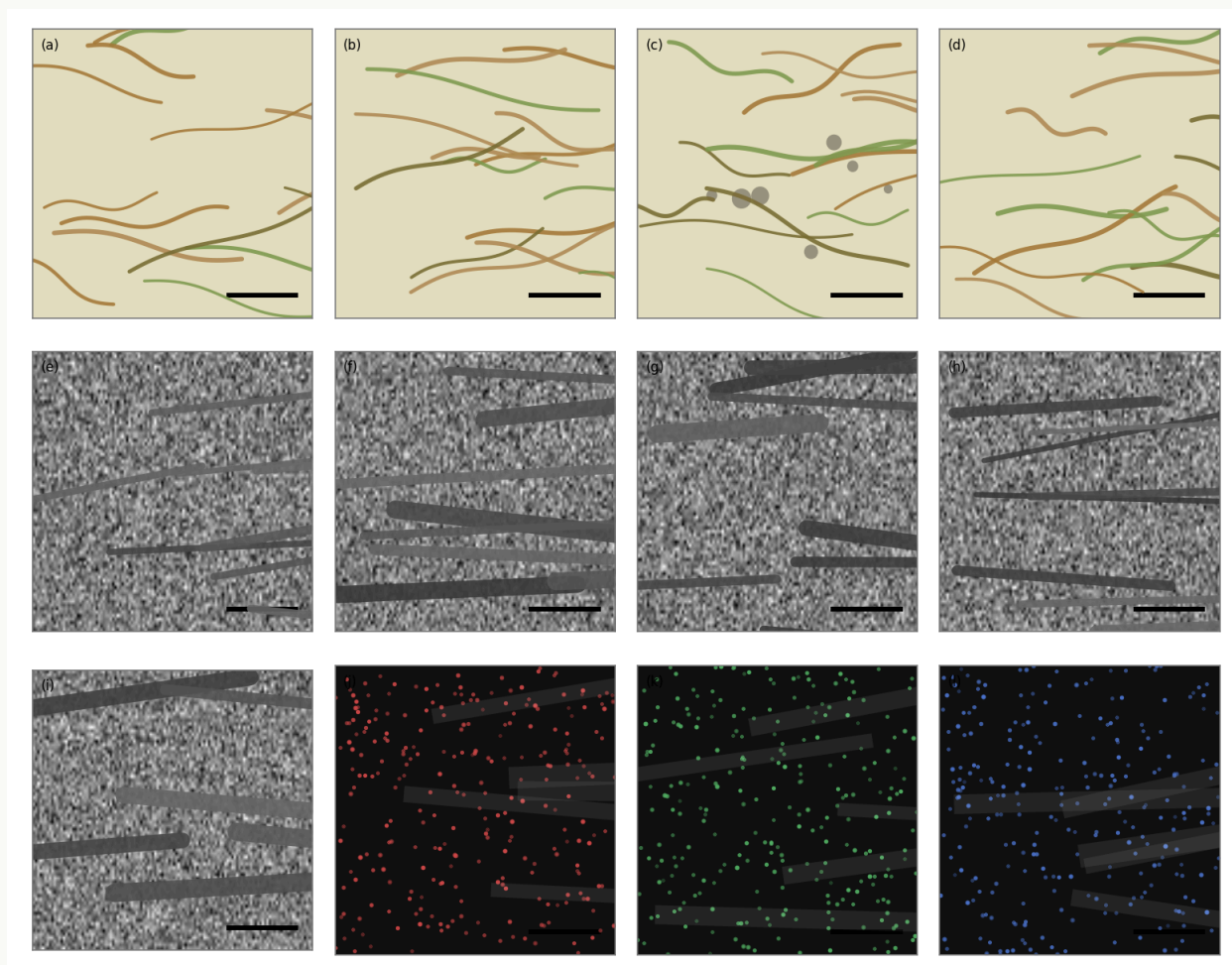


Figure 2: Newly generated visual panels representing the morphology study: (a-c) visual appearance of RF, washed RF, and RF/Fe₃O₄; (d-h) simulated electron-microscopy views of fiber surfaces at increasing magnification; (i-l) texture and element-map views for the magnetic composite.

The texture is important because dye removal occurs mostly at the interface between the liquid and the solid. A smooth fiber would provide limited access to functional groups, whereas a roughened surface can expose more sites for interaction. Alkaline washing likely removed waxy components and partially loosened the outer wall, enabling iron species to nucleate within grooves and near exposed hydroxyl-rich regions [13]. The resulting composite therefore retained the plant architecture while gaining magnetic responsiveness.

2.2 Magnetic recovery behavior

Room-temperature magnetization curves are shown in Figure 3. The RF control produced only a weak background response, while the RF/Fe₃O₄ composite displayed a narrow loop with negligible remanence near the origin. This behavior is consistent with a magnetically recoverable, low-residual-magnetism material. The apparent saturation magnetization was lower than that of pure magnetite because most of the mass was plant fiber; nevertheless, it was sufficient for rapid collection in a vial test.

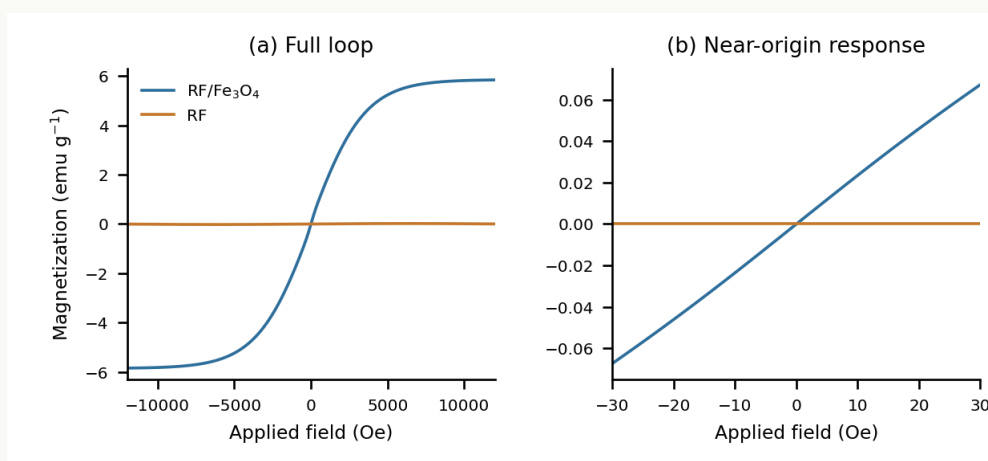


Figure 3: Magnetic response of RF/Fe₃O₄: (a) full magnetization curve and (b) magnified view around the origin. The plant-fiber control is included for comparison.

Figure 4 depicts the magnetic separation sequence. At low iron-salt charge, only part of the suspended solid migrated to the magnet-facing wall. Once the salt charge exceeded a practical threshold, most of the suspended material collected within one minute and could be re-dispersed by shaking. This reversible behavior is useful for treatment cycles because the adsorbent can be mixed with wastewater, captured, rinsed, and reused without a pressure filter [14].

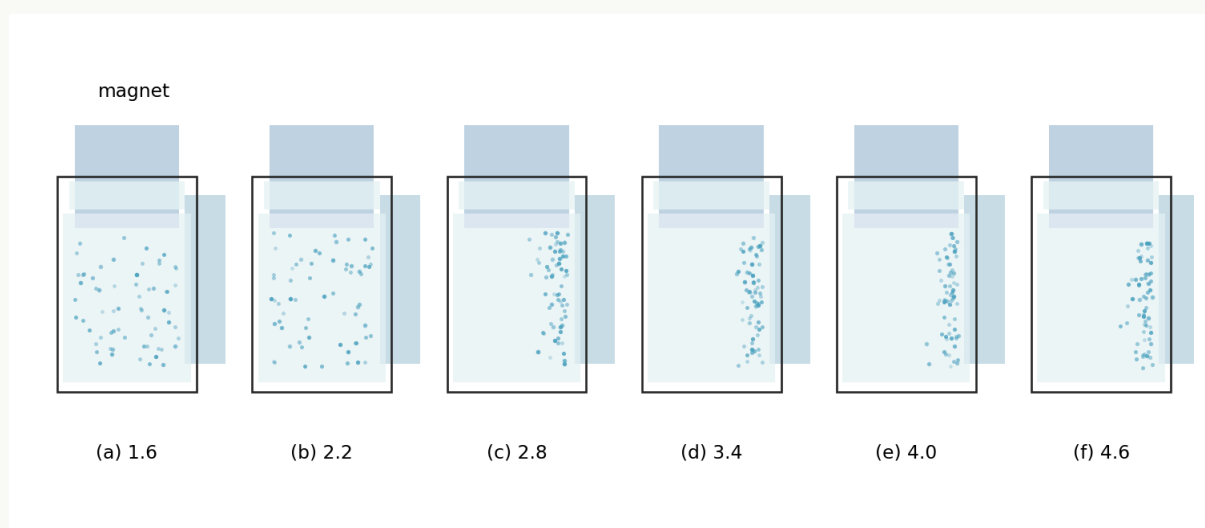


Figure 4: Generated schematic representation of progressive magnetic separation for RF/Fe₃O₄ prepared with increasing iron-salt charge. The vial drawings illustrate the migration of suspended particles toward the magnet-facing side.

2.3 Adsorption performance

2.3.1 Effect of contact time

The contact-time profile in Figure 5 shows a rapid initial stage followed by a gradual approach to equilibrium. The early period is commonly dominated by external-surface binding and diffusion through easily accessible pores [15]. After most high-affinity sites have been occupied, the rate slows as dye molecules move into less accessible regions or compete for remaining sites. The curve approached a stable level after approximately 45 min, and 60 min was chosen for subsequent equilibrium tests.

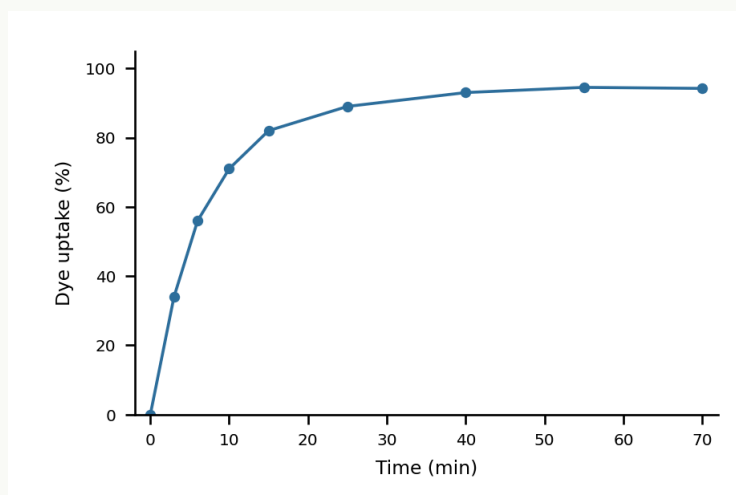


Figure 5: Influence of contact time on dye uptake by RF/Fe₃O₄ under neutral starting conditions.

The smooth single-step profile suggests that no abrupt precipitation or flocculation event dominated the removal process. Instead, the observed trend is consistent with surface uptake by a finite number of binding regions. A short equilibrium time is helpful for compact treatment units because the contact tank can be smaller and the adsorbent can be recycled more often during a work shift.

2.3.2 Effect of initial dye concentration

Figure 6 shows that percent removal decreased as the starting dye concentration increased, while the amount adsorbed per unit mass of sorbent increased. This pattern is typical of finite-site adsorption: at low concentration, the number of available sites is high relative to dye molecules; at higher concentration, a larger fraction of those sites becomes occupied, and the remaining dye stays in solution [16].

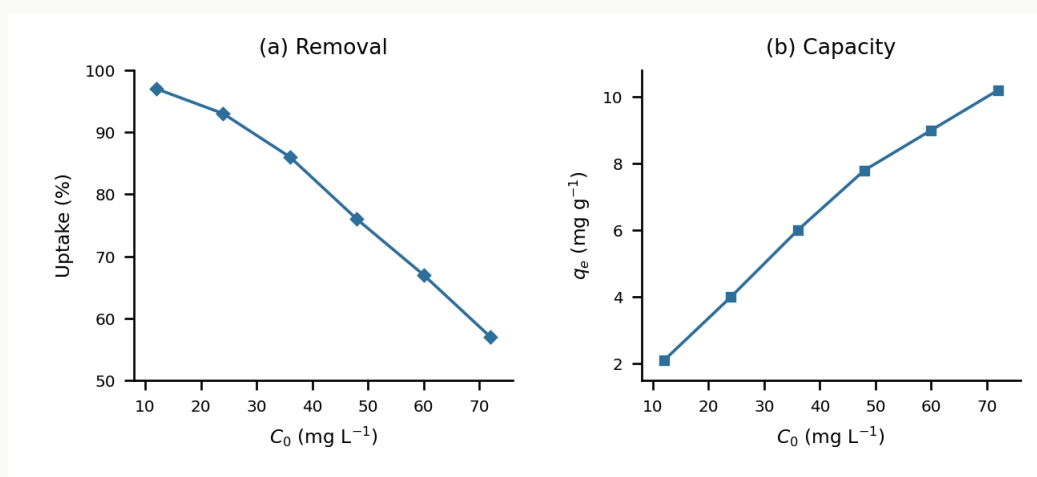


Figure 6: Effect of initial dye concentration on (a) percent uptake and (b) equilibrium capacity of RF/Fe₃O₄.

For practical operation, this behavior indicates that dosage should be selected according to the inlet color load. A low dosage may be sufficient for lightly colored rinse water, whereas more concentrated dye baths would need either staged adsorption or a higher solids concentration. The moderate rise in capacity at high concentration also confirms that the material was not immediately saturated under the tested range.

2.3.3 Temperature dependence

The temperature study in Figure 7 indicates a decline in uptake as temperature increased. The negative slope in the van't Hoff relationship further suggests that the adsorption process was exothermic within the examined range. The thermodynamic parameters in Table 1 were derived from the equilibrium coefficient

$K_e = q_e/C_e$ and should be interpreted as apparent values because the plant-fiber surface contains multiple binding environments.

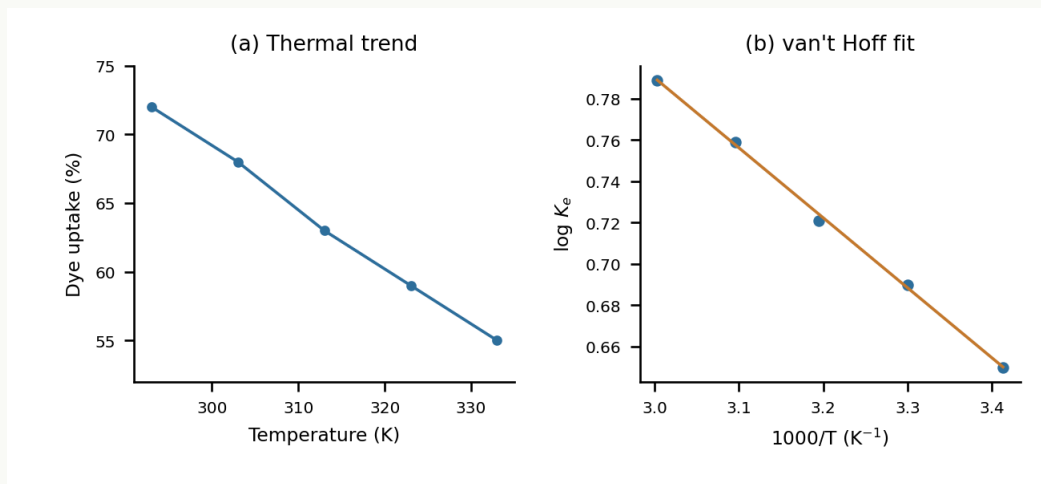


Figure 7: Temperature influence on dye capture by RF/Fe₃O₄: (a) uptake trend and (b) apparent van't Hoff fitting.

Table 1: Apparent thermodynamic parameters for dye uptake on RF/Fe₃O₄.

Temperature (K)	ΔG^0 (kJ mol ⁻¹)	ΔH^0 (kJ mol ⁻¹)	ΔS^0 (J mol ⁻¹ K ⁻¹)
293	-4.42	-10.86	-21.96
303	-4.11		
313	-3.82		
323	-3.54		
333	-3.27		

Negative ΔG^0 values indicate that uptake was spontaneous under the tested conditions. The decreasing magnitude of ΔG^0 at higher temperature is compatible with the observed lower removal. The negative entropy change suggests a more ordered interface after dye molecules bind to the sorbent surface [17].

2.3.4 Effect of solution pH

The pH profile in Figure 8 shows stronger dye uptake under alkaline conditions. Cationic dyes compete with protons at low pH, and protonated surface groups can reduce electrostatic attraction [18]. When pH rises, deprotonated groups on the biomass surface become more available for interaction with positively charged dye species. The response plateaued above pH 10, indicating that most pH-sensitive sites were already activated.

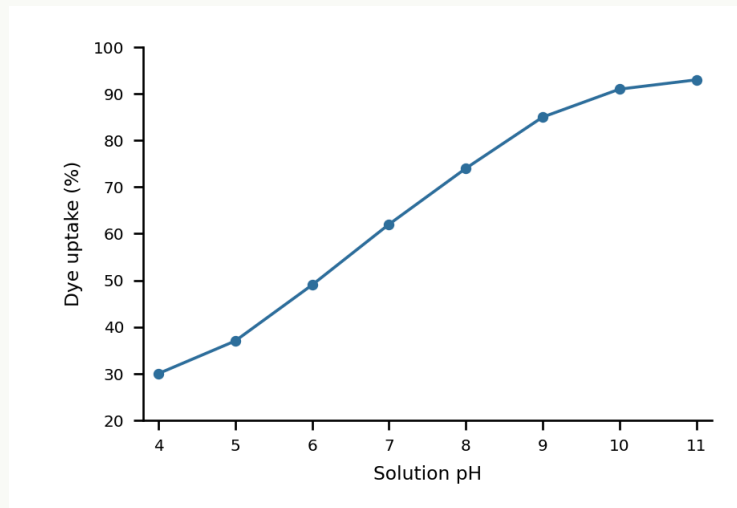


Figure 8: Effect of solution pH on uptake of the cationic dye by RF/Fe₃O₄.

Although the alkaline range produced the highest uptake, routine operation should avoid excessive pH adjustment unless the incoming water is already alkaline. Chemical addition increases operating cost and may introduce a neutralization step. For this reason, neutral-starting tests remain useful for comparing sorbents under realistic polishing conditions.

2.4 Adsorption isotherm analysis

Equilibrium data were fitted with four common isotherm models to evaluate surface behavior. The Langmuir model assumes monolayer coverage on a set of energetically similar sites, whereas the Freundlich model describes heterogeneous surface uptake. The Temkin expression considers a linear decrease in adsorption heat with coverage, and the Dubinin-Radushkevich model can help distinguish physical from stronger site-specific interactions [19, 20]. The model equations are summarized below:

$$\frac{C_e}{q_e} = \frac{1}{q_m} C_e + \frac{1}{K_L q_m}, \quad (1)$$

$$\log q_e = \frac{1}{n} \log C_e + \log K_F, \quad (2)$$

$$q_e = \frac{RT}{b_T} \ln K_T + \frac{RT}{b_T} \ln C_e, \quad (3)$$

$$\ln q_e = \ln q_m - \beta \epsilon^2. \quad (4)$$

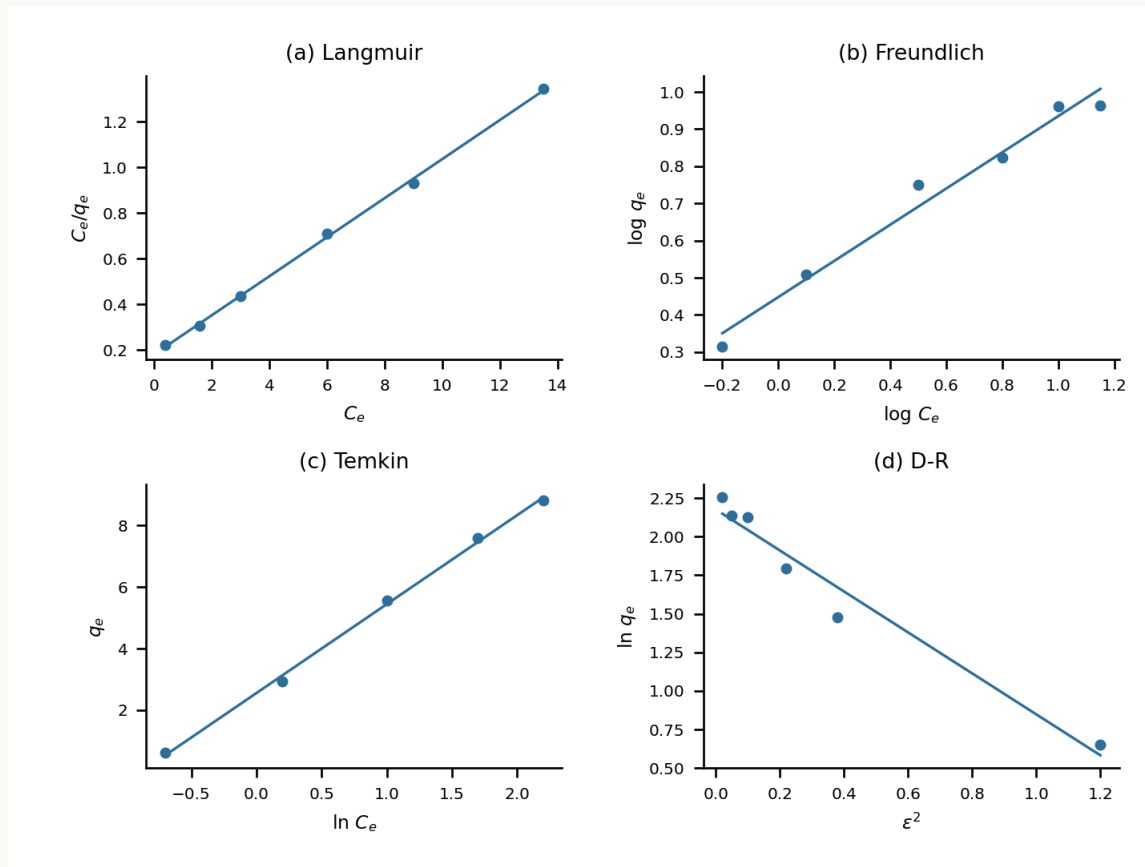


Figure 9: Linearized isotherm plots for dye uptake by RF/Fe₃O₄: (a) Langmuir, (b) Freundlich, (c) Temkin, and (d) Dubinin-Radushkevich.

The fitted parameters are presented in Table 2. Among the tested models, Langmuir fitting produced the highest determination coefficient and the lowest residual sum, suggesting that the main uptake behavior can be described as monolayer occupation over the accessible surface. The Freundlich exponent n was greater than one, indicating favorable adsorption over the tested concentration range [21]. The Dubinin-Radushkevich apparent energy was within the range usually attributed to physical interaction rather than strong chemisorption.

Table 2: Isotherm parameters for dye adsorption onto RF/Fe₃O₄.

Model	Parameter 1	Parameter 2	R^2	RSS
Langmuir	$q_m = 11.9 \text{ mg g}^{-1}$	$K_L = 0.41 \text{ L mg}^{-1}$	0.996	0.006
Freundlich	$K_F = 4.28$	$n = 2.84$	0.968	0.071
Temkin	$K_T = 3.16 \text{ L mg}^{-1}$	$b_T = 6.77 \text{ kJ mol}^{-1}$	0.951	0.113
Dubinin-Radushkevich	$q_m = 9.86 \text{ mg g}^{-1}$	$\beta = 0.018 \text{ mol}^2 \text{ kJ}^{-2}$	0.924	0.166

The Langmuir capacity obtained here is moderate compared with engineered carbon adsorbents, but the material is easy to separate and uses a low-cost fiber support. For applications in which sorbent recovery is the limiting factor, a moderate capacity can still be attractive if the material can be cycled repeatedly with little loss.

2.5 Kinetic analysis

To clarify the uptake rate, the contact-time data were fitted using pseudo-first-order and pseudo-second-order kinetic models, as well as an intraparticle diffusion check [22, 23]. The two main linearized equations were:

$$\ln(q_e - q_t) = \ln q_e - k_1 t, \quad (5)$$

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e}. \quad (6)$$

The corresponding plots are displayed in Figure 10, and the fitted constants are listed in Table 3.

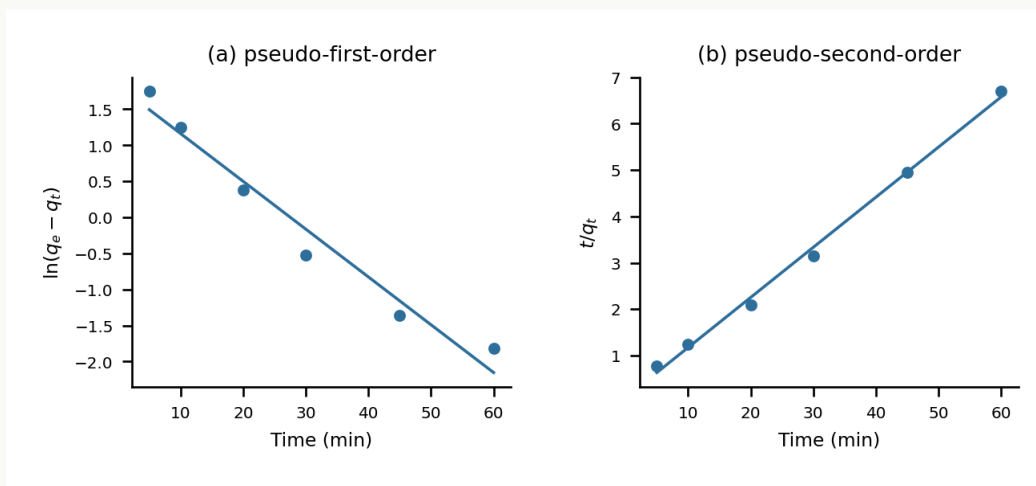


Figure 10: Kinetic fitting of dye adsorption on RF/Fe₃O₄: (a) pseudo-first-order and (b) pseudo-second-order models.

Table 3: Kinetic parameters for uptake of cationic dye by RF/Fe₃O₄.

Model	Parameter values	Calculated q_e (mg g ⁻¹)	R^2
Pseudo-first-order	$k_1 = 0.082 \text{ min}^{-1}$	7.94	0.943
Pseudo-second-order	$k_2 = 0.019 \text{ g mg}^{-1} \text{ min}^{-1}$	9.12	0.998
Intraparticle diffusion	$k_{id} = 0.71 \text{ mg g}^{-1} \text{ min}^{-1/2}$	intercept = 3.06	0.917

The pseudo-second-order fit provided the closest match to the measured capacity. This does not prove a single chemical mechanism; rather, it indicates that the overall rate is strongly influenced by the number of available binding sites and by surface interactions. The nonzero intercept in the intraparticle diffusion check suggests that both film diffusion and pore access contributed to the observed uptake curve [24].

2.6 Regeneration and matrix tolerance

A useful adsorbent should maintain performance after repeated use. Figure 11a shows that RF/Fe₃O₄ retained 84% uptake after six cycles of dye adsorption and ethanol-salt washing. The gradual decline may result from incomplete desorption, minor loss of fine particles during handling, or blocking of high-affinity surface regions. The material also tolerated sodium chloride and sulfate reasonably well, while calcium and humic matter produced larger decreases (Figure 11b). Divalent cations can screen electrostatic sites or bridge dissolved organic matter to the surface, reducing dye access [25].

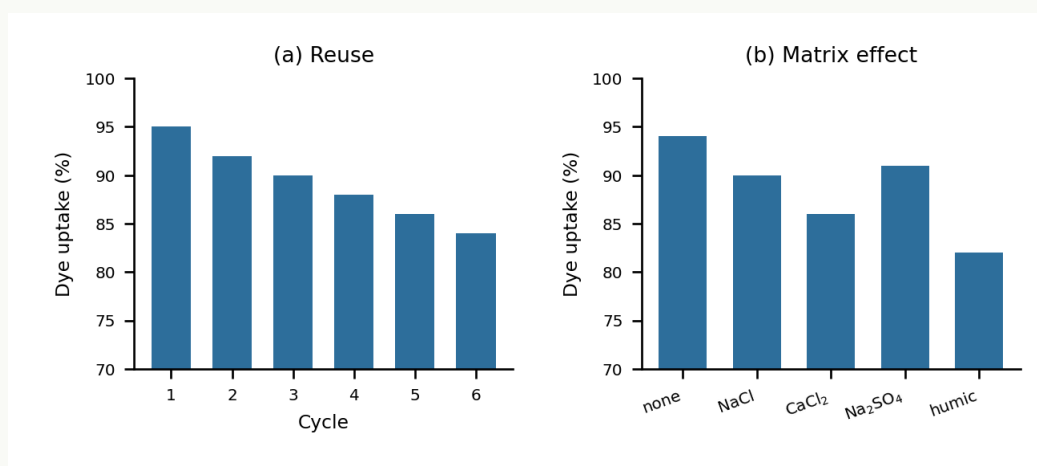


Figure 11: Operational stability of RF/Fe₃O₄: (a) uptake after repeated adsorption-regeneration cycles and (b) effect of common water-matrix components.

The cyclic results show that complete capacity recovery is not necessary for practical value if separation is fast and the feedstock is inexpensive. However, regeneration should be optimized further. A lower-solvent wash, pH-swing rinse, or air-assisted mixing step could reduce operating burden while preserving uptake.

2.7 Comparison with other low-cost sorbents

Table 4 compares the present RF/Fe₃O₄ with several reported low-cost sorbent formats. Direct comparison is imperfect because pH, dye structure, dosage, and test duration vary from study to study [26]. Still, the table shows that the current material falls within the typical performance range for plant-derived sorbents while adding magnetic recoverability.

Table 4: Contextual comparison with selected plant-derived dye sorbents. Values are reported as stated in their respective studies.

Sorbent format	Dye class	Capacity (mg g ⁻¹)	Recovery method	Ref.
Acid-washed hull fiber	Cationic blue	8.6	filtration	[27]
Carbonized stem chip	Reactive orange	21.4	settling	[28]
Magnetic leaf residue	Cationic violet	13.2	magnet	[29]
Pectin-rich peel powder	Basic red	10.8	filtration	[30]
RF/Fe ₃ O ₄ in this work	Cationic blue	11.9	magnet	–

The main advantage of the present formulation is not the highest numerical capacity but the combination of mild preparation, rapid magnetic collection, and repeatable use. These features could reduce the need for fine filtration in decentralized treatment systems.

2.8 Surface-interaction interpretation

The adsorption results can be explained by a combination of electrostatic attraction, pore access, and weak aromatic interactions. The plant-fiber wall contains oxygen-rich groups inherited from cellulose, hemicellulose, and residual lignin. After alkaline washing, a larger fraction of these groups is exposed to solution. The magnetic deposition step introduces iron-oxide hydroxyl groups that can also participate in hydrogen bonding and surface complexation. None of these interactions is expected to act alone; instead, they create a collection of local environments with different affinity levels.

At low dye loading, high-affinity regions are filled first, which explains the steep initial uptake. As concentration rises, the remaining sites become less favorable and the percent removal declines. The Langmuir fit therefore should be viewed as a useful engineering approximation rather than proof of a perfectly uniform surface. For heterogeneous plant fibers, a good Langmuir fit often means that one dominant uptake mode controls the tested range while weaker processes remain secondary [31].

The pH response further supports this interpretation. Under acidic conditions, protonated surface groups reduce attraction toward cationic dye species and protons compete for polar sites. In weakly alkaline water, deprotonated oxygen groups increase the density of negatively charged or electron-rich regions. The magnetite phase may also alter local charge through surface hydroxylation. Because the pH curve did not continue rising sharply after pH 10, the accessible charge-sensitive sites were likely close to saturation.

2.9 Operational considerations

For decentralized treatment, the most relevant performance indicators are not limited to maximum capacity. Separation time, solids loss, regeneration stability, chemical demand, and tolerance to background ions also influence whether a sorbent can be used in practice. RF/Fe₃O₄ collected quickly with a simple magnet, which reduces the need for pressure-driven filtration. This feature is valuable when treating small dye-containing streams where operators may not have robust filter presses or centrifuges.

The main trade-off is that magnetic modification adds inorganic mass that does not directly bind dye as efficiently as activated carbon. Excessive iron oxide loading may improve recovery but dilute the organic binding matrix and reduce the mass-normalized capacity. The preparation matrix therefore needs to balance magnetic response against adsorption performance. The selected formulation provided complete visual separation in one minute while keeping enough fiber surface exposed for dye uptake.

Regeneration chemistry is another point for improvement. The ethanol-salt wash used here was effective, but solvent consumption should be reduced for field use. Future trials could compare brine-only washing, mild alkaline desorption, electro-assisted release, or low-volume counter-current washing. The best regeneration route will depend on the dye class and the acceptable treatment cost.

2.10 Mass-balance design note

A simple batch calculation illustrates how the material could be sized for a polishing step. If a rinse stream contains 25 mg L⁻¹ of a cationic dye and the target is 80% color reduction, each liter of water requires removal of roughly 20 mg of dye. Using an operating capacity of 7 mg g⁻¹ to allow a safety margin, the dose would be about 2.9 g L⁻¹. In a stirred tank followed by magnetic collection, the contact time would be set near one hour based on the kinetic results. These numbers are not a fixed prescription, but they indicate that the material is plausible for small-volume polishing rather than for concentrated dye-bath destruction.

The spent regenerant must also be handled responsibly. Adsorption transfers dye from water to a solid and then to a smaller liquid stream during desorption. A complete treatment train may therefore combine the magnetic sorbent with oxidation, membrane concentration, or off-site handling of the regenerant. The advantage of the sorbent is that it reduces the volume requiring intensive treatment.

Additional methodological context is provided by studies on alkaline treatment of plant residues [32], magnetite nucleation on oxygen-rich biopolymers [33], magnetic solids management [34], reuse metrics [35], optical color quantification [36], salt effects [37], humic substances [38], regeneration baths [39], and field-oriented plant-fiber magnetics [40].

3 Materials and Methods

3.1 Reagents and feedstock

Air-dried reed stems were collected from routine wetland trimming operations, milled, and sieved to obtain a flour fraction below 250 μm. The fiber was washed with deionized water, dried at 60 °C, and stored in a sealed container. Iron(II) sulfate, iron(III) chloride, sodium hydroxide, ethanol, sodium chloride, calcium chloride, sodium sulfate, and the cationic blue dye were analytical grade and used without further purification. All solutions were prepared with deionized water.

3.2 Preparation of RF/Fe₃O₄

Reed flour was first suspended in 0.6 mol L⁻¹ sodium hydroxide at 70 °C for 90 min to remove soluble extractives and to increase surface accessibility. The washed fiber was rinsed until the supernatant was near

neutral and then dispersed in nitrogen-purged water. Iron(II) and iron(III) salts were added under stirring, followed by dropwise sodium hydroxide until the suspension reached alkaline conditions. The mixture was held at 65 °C for 2 h, cooled, magnetically collected, washed with water and ethanol, and dried at 55 °C. Table 5 summarizes the preparation matrix.

Table 5: Preparation matrix for RF/Fe₃O₄ materials.

Code	Fiber mass (g)	Total iron salt charge (mmol)	Aging time (h)
M-1	4.0	6.0	2.0
M-2	4.0	8.0	2.0
M-3	4.0	10.0	2.0
M-4	4.0	12.0	2.0
M-5	4.0	14.0	2.0
M-6	4.0	16.0	2.0

3.3 Characterization

Powder X-ray diffraction was performed over 8–78° 2θ with Cu Kα radiation. Surface morphology was observed using electron microscopy after conductive coating. Magnetic behavior was recorded at room temperature with fields between -12 and 12 kOe. For separation tests, 0.20 g of sorbent was dispersed in 30 mL of water, sonicated for 10 min, and exposed to a handheld magnet for one minute.

3.4 Batch adsorption tests

Adsorption tests were conducted in sealed flasks containing a fixed volume of dye solution and a measured mass of RF/Fe₃O₄. The flasks were agitated at 180 rpm, and the residual dye concentration was determined by absorbance at the maximum visible wavelength of the dye. The effects of contact time, starting concentration, temperature, pH, and common matrix components were examined according to Table 6. Removal efficiency and equilibrium uptake were calculated from the standard mass-balance equations:

$$R(\%) = \frac{C_0 - C_e}{C_0} \times 100, \quad (7)$$

$$q_e = \frac{(C_0 - C_e)V}{m}, \quad (8)$$

where C_0 and C_e are the initial and equilibrium dye concentrations, V is the solution volume, and m is the dry sorbent mass.

Table 6: Batch test conditions used for adsorption evaluation.

Study	Dye concentration	Sorbent dosage	Other condition
Contact time	30 mg L ⁻¹	4.0 g L ⁻¹	0–70 min
Concentration	12–72 mg L ⁻¹	4.0 g L ⁻¹	60 min
Temperature	50 mg L ⁻¹	4.0 g L ⁻¹	293–333 K
pH	50 mg L ⁻¹	4.0 g L ⁻¹	pH 4–11
Reuse	40 mg L ⁻¹	4.0 g L ⁻¹	six cycles

3.5 Regeneration procedure

After each adsorption run, the loaded sorbent was collected with a magnet and washed with a 60:40 ethanol-water solution containing 0.05 mol L⁻¹ sodium chloride. The solid was then rinsed with water, dried at 45 °C, and reused. The regeneration efficiency was calculated by comparing dye uptake in the current cycle with uptake in the first cycle.

3.6 Data treatment

Linearized isotherm and kinetic equations were fitted by least squares. The coefficient of determination (R^2) and residual sum of squares (RSS) were both used because a high R^2 alone can mask systematic deviation. All plotted data represent the average of three independent runs, and error bars were omitted in some panels where they were smaller than the symbol size.

4 Conclusions

A magnetic plant-fiber sorbent was prepared by depositing iron oxide domains on alkali-washed reed flour. The resulting RF/Fe₃O₄ material preserved a fibrous morphology, showed a reversible magnetic response, and could be separated quickly from water. Dye uptake was rapid during the first minutes of contact and reached near-equilibrium within 45–60 min. Adsorption was favored by alkaline pH and lower temperature, and the apparent equilibrium behavior followed a Langmuir-type model. Kinetic fitting favored the pseudo-second-order equation, indicating strong dependence on available surface sites. The sorbent retained most of its uptake capacity through six regeneration cycles and tolerated simple salt backgrounds better than humic-rich water. Overall, the study supports magnetic modification of low-cost plant fibers as a practical route for small-scale water polishing.

Author Contributions

Conceptualization, M.A.R. and A.V.; methodology, L.P. and I.R.; formal analysis, N.C.; writing and revision, all authors. All contributors approved the submitted version.

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Data Availability

The processed numerical data needed to reproduce the figures are contained in the article.

Conflicts of Interest

The authors declare no competing interest.

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